Hessian Lipschitzness Does Not Rescue the Dimension-Free Logarithmic Local Rate

July 14, 2026

Abstract

We give a rigorous option-(B) answer to the Hessian-Lipschitz local-rate question. A two-scale radial-softplus construction produces smooth genuine Hessians, already in equilibrium-whitened coordinates, for which

$$\kappa \asymp \frac{d}{\sigma^2}, \qquad \gamma_{\mathrm{loc}} \asymp \frac{\sigma}{\sqrt{d}} \asymp \kappa^{-1/2}, \qquad L_H \asymp \frac{\sqrt{d}}{\sigma^2}, \qquad \tau_H^2 \gtrsim \frac{\sqrt{d}}{\sigma} \asymp \sqrt{\kappa}.$$

Taking $(d,\sigma) = (n^4, n)$ gives $L_H \asymp 1$; taking $(d,\sigma) = (n^6, n^2)$ gives $L_H \asymp n^{-1} \to 0$. In both cases $\kappa \asymp n^2$ and $\gamma_{\mathrm{loc}} \asymp n^{-1} \asymp \kappa^{-1/2}$. Thus even a vanishing whitened Hessian-Lipschitz modulus does not rescue any polylogarithmic local rate. The proof uses exact Gaussian isotropization and an explicit affine Rayleigh witness. No Frobenius/operator conversion, symmetry-only surrogate, or noncommutative Loewner transfer occurs.

We also complete the requested calibration. The attached unit-scale counterexample has $L_H \asymp \sqrt{\kappa}$. The fixed-width logistic smoothing of the one-dimensional rare-tail step has $L_H \asymp \sqrt{\kappa}$ but is only preasymptotically log-sharp; a genuine asymptotically log-sharp compact-tail smoothing exists with $L_H = O(\sqrt{\kappa})$. Across unrestricted smoothings there is no canonical minimal power, although bounded or polylogarithmic L_H cannot realize the one-dimensional logarithmic extremizer.

1 Main result and norm convention

For a C^3 potential define its Hessian operator-Lipschitz modulus by

$$L_H(V) := \sup_{x \neq y} \frac{\|\nabla^2 V(x) - \nabla^2 V(y)\|_{\mathrm{op}}}{\|x-y\|}.$$

Since the third derivative is symmetric,

$$L_H(V) = \sup_x \sup_{\|h\|=\|u\|=1} |\nabla^3 V(x)[h,u,u]|. \tag{1.1}$$

Indeed, the right side is $\sup_{x,\|h\|=1} \|D(\nabla^2 V)(x)[h]\|_{\mathrm{op}}$; the fundamental theorem of calculus proves one inequality and directional difference quotients prove the other. Thus every L_H below is the tensor/operator quantity in (H2), not a Frobenius norm. For a symmetric trilinear form $\mathcal{T}$, we also write $\|\mathcal{T}\|_{\mathrm{inj}} := \sup_{\|h\|=\|u\|=\|v\|=1} |\mathcal{T}[h,u,v]|$.

Theorem 1.1 (Vanishing-L_H square-root counterexample)**.** *There are numerical constants $c, C > 0$ and $n_0 \in \mathbb{N}$ such that for every integer $n \geq n_0$ there is a C^∞, strongly convex potential $V_n : \mathbb{R}^{n^6+1} \to \mathbb{R}$ satisfying, for $Z \sim \mathcal{N}(0, I_{n^6+1})$,*

$$\mathbb{E}\nabla V_n(Z) = 0, \qquad \mathbb{E}\nabla^2 V_n(Z) = I. \tag{1.2}$$

If α_n, β_n are its optimal scalar curvature constants and $\kappa_n = \beta_n/\alpha_n$, then

$$\frac{c}{n} \leq \alpha_n \leq \frac{C}{n}, \qquad cn \leq \beta_n \leq Cn, \qquad cn^2 \leq \kappa_n \leq Cn^2. \tag{1.3}$$

Its optimal whitened Hessian-Lipschitz modulus and linearized gap obey

$$\frac{c}{n} \leq L_H(V_n) \leq \frac{C}{n}, \qquad \frac{c}{n} \leq \gamma_{\mathrm{loc}}(V_n) \leq \frac{C}{n}, \tag{1.4}$$

and its first-Hermite Hessian coupling satisfies

$$\tau_H(V_n)^2 \geq cn. \tag{1.5}$$

Consequently

$$L_H(V_n) \longrightarrow 0, \qquad \gamma_{\mathrm{loc}}(V_n) \asymp \kappa_n^{-1/2}, \qquad \gamma_{\mathrm{loc}}(V_n)^{-1} \asymp \sqrt{\kappa_n} \gg \log \kappa_n.$$

The same construction with transverse dimension $d = n^4$ and scale $\sigma = n$ has $L_H \asymp 1$, $\kappa \asymp n^2$, and $\gamma_{\mathrm{loc}} \asymp n^{-1}$. Hence the conjectured bound fails even at a fixed Hessian-Lipschitz modulus, independently of how a candidate Φ is defined near zero.

The construction and proof occupy Sections 3–7. We first perform the mandatory calibration.

2 Mandatory calibration

2.1 The attached unit-scale radial-softplus family

The attached counterexample uses

$$r(y) = \sqrt{1 + \|y\|^2}, \quad F(s) = \log(1 + e^s), \quad U(t, y) = F(r(y) - \sqrt{d} - \delta t),$$

and $V = \alpha\|z\|^2/2 + c_d U + b^\top z$, where $c_d \asymp \sqrt{d}$ and $\kappa_d \asymp d$. The exact third-order chain rule is

$$\nabla^3 U = F'''(s)(\nabla s)^{\otimes 3} + F''(s)\ \mathrm{Sym}(\nabla^2 s \otimes \nabla s) + F'(s)\nabla^3 s. \tag{2.1}$$

Here $\|\nabla s\|$, $\|\nabla^2 s\|_{\mathrm{op}}$, and the injective norm of $\nabla^3 s$ are uniformly bounded, as are the first three derivatives of F. Hence $L_H(U) \leq C$.

For the reverse bound, take $y = \rho e_1$, let $\rho \to \infty$, and choose t so that $s = \log 2$. In the e_1 direction, $\partial_1 r \to 1$ while $\partial_{11} r, \partial_{111} r \to 0$, and

$$F'''(\log 2) = -\frac{2}{27}.$$

Thus $L_H(U) \geq 2/27$. The quadratic and affine parts have zero third derivative, so

$$L_H(V_d) \asymp c_d \asymp \sqrt{d} \asymp \sqrt{\kappa_d}. \tag{2.2}$$

This is the exact whitened tensor/operator scale. Merely widening F does not remove it: the term $F'(s)\nabla^3 r$ in (2.1) remains of unit size. The counterexample below widens the radial core and the outer softplus simultaneously.

2.2 The one-dimensional rare-tail family

Let $K \ge 2$, $a = K^{-1}$, and consider the logistic smoothing

$$W_{K,\varepsilon}(x) = a + (K-a)\ell\left(\frac{x-r}{\varepsilon}\right), \qquad \ell(u) = \frac{1}{1+e^{-u}}, \tag{2.3}$$

where r is chosen so that $\mathbb{E}W_{K,\varepsilon}(Z) = 1$. The normalization is exactly

$$\mathbb{E}\ell((Z-r)/\varepsilon) = \frac{1}{K+1}, \tag{2.4}$$

and

$$L_H(W_{K,\varepsilon}) = \|W'_{K,\varepsilon}\|_\infty = \frac{K-K^{-1}}{4\varepsilon}. \tag{2.5}$$

For fixed $\varepsilon > 0$, this smoothing is not asymptotically log-sharp. For every polynomial P of degree at most two, dominated convergence gives

$$e^{r/\varepsilon}\mathbb{E}[P(Z)\ell((Z-r)/\varepsilon)] \longrightarrow \mathbb{E}[P(Z)e^{Z/\varepsilon}] \quad (r \to \infty). \tag{2.6}$$

Equations (2.4)–(2.6) yield

$$r = \varepsilon \log K + \frac{1}{2\varepsilon} + o(1), \qquad T := \mathbb{E}[ZW_{K,\varepsilon}(Z)] \to \varepsilon^{-1}, \qquad H := \mathbb{E}[(Z^2-1)W_{K,\varepsilon}(Z)] \to \varepsilon^{-2}. \tag{2.7}$$

The determinant of the packed 2×2 generator tends to one and its trace tends to $2 + (2\varepsilon^2)^{-1}$, so its gap tends to a positive ε-dependent constant. Thus the fixed width 0.1 experiment in the audit is preasymptotic. Formula (2.5) nevertheless shows that its Hessian-Lipschitz scale is $\Theta(K) = \Theta(\sqrt{\kappa})$.

A genuinely asymptotically log-sharp smooth realization is easy to obtain, but it still does not have small L_H. Let $h \in C^\infty(\mathbb{R})$ be nondecreasing, $h = 0$ on $(-\infty, 0]$, and $h = 1$ on $[1,\infty)$. Define

$$W_K(x) = K^{-1} + (K - K^{-1})h(x-r), \qquad \mathbb{E}W_K(Z) = 1. \tag{2.8}$$

If r_K is determined by $\mathbb{P}(Z > r_K) = 1/(K+1)$, then $r \le r_K \le r+1$, so $r \asymp r_K \asymp \sqrt{\log K}$. With $\nu_K(dx) = W_K(x)\varphi(x)\,dx$, one has

$$\mathbb{E}_{\nu_K} Z \asymp r_K, \qquad \mathbb{E}_{\nu_K} Z^2 \asymp r_K^2, \qquad 0 \le \mathrm{Var}_{\nu_K}(Z) \le C. \tag{2.9}$$

For completeness, the first two estimates follow by using the support $[r,\infty)$ for the lower bounds and splitting at $2r$ for the upper bounds. For the variance, center at $r+1$: the transition contribution is bounded by its total mass, the tail beyond $r+1$ has a bounded Gaussian overshoot second moment, and the $K^{-1}\varphi$ background contributes $O(r_K^2/K)$. Therefore

$$\frac{1}{2} \le \det L_K = \frac{1+\mathrm{Var}_{\nu_K}(Z)}{2} \le C, \qquad \mathrm{Tr}\, L_K \asymp r_K^2, \qquad \gamma_{\mathrm{loc}}(W_K) \asymp \frac{1}{1+\log\kappa}, \quad \kappa = K^2. \tag{2.10}$$

This fixed-shape smoothing has

$$L_H(W_K) = (K - K^{-1})\|h'\|_\infty = \Theta(K) = \Theta(\sqrt{\kappa}). \tag{2.11}$$

There is no canonical "minimal L_H over all smoothings" unless the smoothing class and the comparison constant in $\gamma_{\mathrm{loc}} \asymp 1/\log\kappa$ are fixed. Here is an explicit reason. Fix $0 < \theta < 1$, put $A = K^\theta$ and $w = (K-A)/A$, and use the cutoff h from (2.8). Choose R_K so large that

$$(K-A)\mathbb{E}[(1+Z^2)h((Z-R_K)/w)] \le K^{-10},$$

and set

$$W_{K,\theta}(x) = K^{-1} + (A - K^{-1})h(x - r) + (K - A)h((x - R_K)/w), \tag{2.12}$$

with r chosen continuously so that $\mathbb{E}W_{K,\theta} = 1$. The two ramps both have slope $O(A)$, the optimal range is $[K^{-1}, K]$, and the remote ramp has negligible mass and first two moments. The main normalization is $\mathbb{E}h(Z - r) = (1 + o(1))/A$, so the same argument as above, with K replaced by A, gives

$$\gamma_{\mathrm{loc}} \asymp (\theta \log K)^{-1}, \qquad L_H \asymp K^{\theta}.$$

Thus different fixed comparison constants give different powers of K.

There is, however, a useful rigorous qualitative lower calibration. If $f \geq 0$, $m = \mathbb{E}f > 0$, and f is L-Lipschitz, then

$$\left|\frac{\mathbb{E}[Zf(Z)]}{m}\right|^2 \leq C \log\left(e + \frac{L}{m}\right). \tag{2.13}$$

To prove it, positivity and Lipschitzness first give $f(0) \leq C(m+L)$ and hence $f(x) \leq C(m+L(1+|x|))$. For the probability density $f\varphi/m$, split its first absolute moment at $R = 2\sqrt{\log(e + L/m)}$; the bulk is at most R, and standard Gaussian tail moments make the remaining contribution universally bounded. Applying (2.13) to $f = W - K^{-1}$ gives a sharp qualitative conclusion. Indeed, let $\mu_2 = \mathbb{E}[W(Z)Z^2]$ and let $\gamma \leq \Lambda$ be the two eigenvalues of the packed generator. Since $\det L = (1 + \mathrm{Var}_{W\varphi} Z)/2 \geq 1/2$, $\Lambda \geq (2\gamma)^{-1}$. Because $\operatorname{Tr} L = (3 + \mu_2)/2 = \gamma + \Lambda$, one has $\mu_2 \geq (2\gamma)^{-1}$ whenever γ is sufficiently small. The characteristic equation then gives

$$T^2 = (1 - \gamma)(1 + \mu_2 - 2\gamma) \geq \frac{1}{4\gamma}.$$

Thus $\gamma \leq C_0/(1+\log \kappa)$ implies $T^2 \geq c(C_0)(1+\log \kappa)$. Since $m = \mathbb{E}(W - K^{-1}) = 1 - K^{-1} \in [1/2, 1]$, inequality (2.13) shows that any uniformly log-sharp family of this normalization must have $L_H \geq \kappa^{c_0}$ for some constant $c_0 > 0$ determined by the gap comparison constant. In particular, the one-dimensional logarithmic extremizer cannot be realized with bounded or polylogarithmic L_H. This corrects the small-L_H premise in the proposed calibration; the high-dimensional construction below is strictly more adverse.

3 The two-scale radial-softplus family

Let d and σ satisfy

$$\sigma \geq 1, \qquad d \geq C_0 \sigma^2, \tag{3.1}$$

where C_0 is a sufficiently large numerical constant. Write $z = (t, y) \in \mathbb{R} \times \mathbb{R}^d$ and set

$$r(y) := \sqrt{\sigma^2 + \|y\|^2}, \qquad R := \sqrt{\sigma^2 + d}, \qquad F_\sigma(s) := \sigma \log(1 + e^{s/\sigma}). \tag{3.2}$$

Put

$$p(s) := F_\sigma'(s), \qquad q(s) := F_\sigma''(s) = \frac{1}{\sigma} p(s)(1 - p(s)).$$

Thus $0 < p < 1$, $0 < q \leq (4\sigma)^{-1}$, and $|F_\sigma'''| \leq (4\sigma^2)^{-1}$.

For $0 \leq \delta \leq 1$, define

$$s_\delta(t, y) := r(y) - R - \delta t, \qquad U_\delta(t, y) := F_\sigma(s_\delta(t, y)), \qquad A_\delta := \nabla^2 U_\delta. \tag{3.3}$$

Writing $S = \|y\|^2$,

$$v_\delta := \nabla s_\delta = (-\delta, y/r), \qquad D(y) := \nabla_y^2 r = \frac{I_d}{r} - \frac{yy^\top}{r^3}. \tag{3.4}$$

The exact Hessian is

$$A_\delta = q(s_\delta)v_\delta v_\delta^\top + p(s_\delta)\begin{pmatrix} 0 & 0 \\ 0 & D(y) \end{pmatrix}. \tag{3.5}$$

Both terms are positive semidefinite. The tangential eigenvalue of D is r^{-1} and its radial eigenvalue is $\sigma^2 r^{-3}$. Since $\|v_\delta\|^2 \le 2$,

$$0 \preceq A_\delta \preceq \frac{3}{2\sigma} I_{d+1}. \tag{3.6}$$

4 Exact Gaussian isotropization

Let $T \sim \mathcal{N}(0,1)$ and $Y \sim \mathcal{N}(0, I_d)$ be independent. In the following expectations $s_\delta = s_\delta(T, Y)$ and $r = r(Y)$. Define

$$Q_{d,\sigma}(\delta) := \mathbb{E}q(s_\delta), \tag{4.1}$$

$$M_{d,\sigma}(\delta) := \frac{1}{d}\mathbb{E}\left[q(s_\delta)\frac{S}{r^2} + p(s_\delta)\left(\frac{d-1}{r} + \frac{\sigma^2}{r^3}\right)\right]. \tag{4.2}$$

Rotation invariance and oddness of the mixed block give

$$\mathbb{E}A_\delta = \operatorname{diag}(\delta^2 Q_{d,\sigma}(\delta), M_{d,\sigma}(\delta) I_d). \tag{4.3}$$

Lemma 4.1 (Uniform scalar estimates and the isotropizing root)**.** *Under* (3.1)*, uniformly for* $0 \le \delta \le 1$*,*

$$\frac{c}{\sigma} \le Q_{d,\sigma}(\delta) \le \frac{1}{4\sigma}, \qquad \frac{c}{R} \le M_{d,\sigma}(\delta) \le \frac{C}{R}. \tag{4.4}$$

If C_0 is large enough, the continuous equation

$$\delta^2 Q_{d,\sigma}(\delta) = M_{d,\sigma}(\delta) \tag{4.5}$$

has a root $\delta_{d,\sigma} \in (0,1)$. For any such root, with $m_{d,\sigma}$ denoting the common value,

$$\delta_{d,\sigma}^2 \asymp \frac{\sigma}{R}, \qquad m_{d,\sigma} \asymp \frac{1}{R}, \qquad \mathbb{E}A_{\delta_{d,\sigma}} = m_{d,\sigma} I_{d+1}. \tag{4.6}$$

Proof. On the event

$$\mathcal{E}_d := \{|S - d| \le 2\sqrt{d},\ |T| \le 1\},$$

Chebyshev's inequality and independence give $\mathbb{P}(\mathcal{E}_d) \ge c_0 > 0$. Moreover

$$|r - R| = \frac{|S-d|}{r+R} \le 2, \qquad |s_\delta| \le 3.$$

Since $\sigma \ge 1$, both $p(s_\delta)$ and $\sigma q(s_\delta)$ are bounded below on this event. This proves the lower bound for Q. On the same event,

$$\operatorname{Tr} D = \frac{d-1}{r} + \frac{\sigma^2}{r^3} \ge c\frac{d}{R},$$

which proves the lower bound for M.

The upper bound for Q is immediate. For $d > 2$,

$$\mathbb{E}\frac{1}{r} \leq \mathbb{E}S^{-1/2} \leq (\mathbb{E}S^{-1})^{1/2} = \frac{1}{\sqrt{d-2}} \leq \frac{C}{R}. \tag{4.7}$$

The first summand in (4.2) is at most $(4\sigma d)^{-1}$, and the second is at most $(1 + d^{-1})\mathbb{E}r^{-1}$. This proves (4.4).

Continuity follows by dominated convergence. The left side minus the right side of (4.5) is negative at zero and positive at one once R/σ is a sufficiently large numerical constant. Hence a root exists. At any root, (4.4) gives (4.6), and (4.3) gives exact isotropy. □

Fix a root, abbreviate $U = U_{\delta_{d,\sigma}}$, $A = \nabla^2 U$, and set

$$\alpha := \frac{\sigma}{R}, \qquad c_{d,\sigma} := \frac{1-\alpha}{m_{d,\sigma}} \asymp R. \tag{4.8}$$

Define

$$V_{d,\sigma}(z) := \frac{\alpha}{2}\|z\|^2 + c_{d,\sigma}U(z) + b_{d,\sigma}^{\top} z, \qquad b_{d,\sigma} := -c_{d,\sigma}\mathbb{E}\nabla U(Z). \tag{4.9}$$

The gradient of U is bounded, so $b_{d,\sigma}$ is finite. It is a multiple of the t direction. The Hessian

$$W_{d,\sigma} = \nabla^2 V_{d,\sigma} = \alpha I + c_{d,\sigma}A \tag{4.10}$$

satisfies, exactly,

$$\mathbb{E}\nabla V_{d,\sigma}(Z) = 0, \qquad \mathbb{E}W_{d,\sigma}(Z) = I. \tag{4.11}$$

The optimal lower curvature is α, since $A(t, y) \to 0$ as $t \to +\infty$ for fixed y. Equation (3.6) gives the upper curvature bound. Conversely, at $y = 0$ and with t chosen so that $s_\delta = 0$, the transverse block of A is $I_d/(2\sigma)$. Thus the optimal upper curvature β satisfies

$$\alpha + \frac{c_{d,\sigma}}{2\sigma} \leq \beta \leq \alpha + \frac{3c_{d,\sigma}}{2\sigma}, \qquad \beta \asymp \frac{R}{\sigma}, \qquad \kappa := \frac{\beta}{\alpha} \asymp \frac{R^2}{\sigma^2} \asymp \frac{d}{\sigma^2}. \tag{4.12}$$

The potential is C^∞ and α-strongly convex, hence normalizable. It is a genuine Hessian and its complete compatibility hierarchy holds classically.

5 The exact Hessian-Lipschitz scale

The third derivative of r is

$$\nabla^3 r[h, u, v] = -\frac{(y\cdot h)(u\cdot v) + (y\cdot u)(h\cdot v) + (y\cdot v)(h\cdot u)}{r^3} \tag{5.1}$$

$$+ 3\frac{(y\cdot h)(y\cdot u)(y\cdot v)}{r^5}. \tag{5.2}$$

Consequently

$$\|\nabla s_\delta\| \leq \sqrt{2}, \qquad \|\nabla^2 s_\delta\|_{\mathrm{op}} \leq \sigma^{-1}, \qquad \|\nabla^3 s_\delta\|_{\mathrm{inj}} \leq 6\sigma^{-2}. \tag{5.3}$$

The exact chain rule is

$$\nabla^3 U[h, u, v] = F_\sigma'''(s)\, ds[h]ds[u]ds[v] \tag{5.4}$$

$$+ q(s)\{\nabla^2 s[h, u]ds[v] + \nabla^2 s[h, v]ds[u] + \nabla^2 s[u, v]ds[h]\} \tag{5.5}$$

$$+ p(s)\nabla^3 s[h, u, v]. \tag{5.6}$$

Equations (5.3)–(5.4) give

$$\sup_z \|\nabla^3 U(z)\|_{\mathrm{inj}} \le \frac{C}{\sigma^2}. \tag{5.7}$$

For the reverse bound, fix $y = (\sigma/2)e_1$ and send $t \to -\infty$. Then $p(s) \to 1$ and $q(s), F_\sigma'''(s) \to 0$, so $\nabla^3 U \to \nabla^3 r$. Directly from (5.1),

$$|\nabla^3 r[e_1, e_1, e_1]| = \frac{c_1}{\sigma^2}$$

for a numerical $c_1 > 0$. Since the quadratic and affine terms in (4.9) have zero third derivative,

$$L_H(V_{d,\sigma}) \asymp \frac{c_{d,\sigma}}{\sigma^2} \asymp \frac{R}{\sigma^2} \asymp \frac{\sqrt{d}}{\sigma^2}. \tag{5.8}$$

This is a direct injective-tensor estimate; no dimension-dependent Frobenius conversion is present.

6 Compatible Rayleigh identity and the inverse gap

If X is the physical covariance perturbation, its packed coordinate is $X/\sqrt{2}$ and its Fisher norm is

$$\|(u, X)\|_\star^2 := \|u\|^2 + \frac{1}{2}\|X\|_F^2. \tag{6.1}$$

Lemma 6.1 (Compatible Bochner–Rayleigh identity)**.** *Let $V \in C^4$ and suppose that its Hessian $W = \nabla^2 V$ and the first two derivatives of W have at most polynomial growth. If $\mathbb{E}W = I$, then the packed generator in the question satisfies*

$$4\mathcal{Q}_W(u, X) = \mathbb{E}[(2u + XZ)^\top W(Z)(2u + XZ)] + \|X\|_F^2. \tag{6.2}$$

If $W \succeq \alpha I$ and $0 < \alpha \le 1$, then

$$\mathcal{Q}_W(u, X) \ge \alpha \|(u, X)\|_\star^2, \qquad \gamma_{\mathrm{loc}}(W) \ge \alpha. \tag{6.3}$$

Proof. Expanding the displayed block operator in the packed coordinate gives

$$\mathcal{Q}_W(u, X) = \|u\|^2 + u^\top T[X] + \frac{1}{2}\|X\|_F^2 + \frac{1}{4}\langle X, H[X]\rangle_F.$$

Gaussian integration by parts and third-derivative symmetry give

$$\mathbb{E}[u^\top W X Z] = u^\top T[X].$$

In components, double Gaussian integration by parts gives

$$\mathbb{E}[Z_a Z_b W_{ij}(Z)] = \delta_{ab}\delta_{ij} + \mathbb{E}[\partial_{ab} W_{ij}(Z)].$$

Full fourth-derivative symmetry then gives

$$X_{ia} X_{jb} \mathbb{E}[\partial_{ab} W_{ij}] = X_{ij} X_{ab} \mathbb{E}[\partial_{ab} W_{ij}],$$

and therefore

$$\mathbb{E}[(XZ)^\top W (XZ)] = \|X\|_F^2 + \langle X, H[X]\rangle_F.$$

Substitution proves (6.2). The displayed family has a bounded Hessian and bounded derivatives of the Hessian of every order because $r \ge \sigma > 0$, so all steps are classical. Finally,

$$4\mathcal{Q}_W(u, X) \ge 4\alpha\|u\|^2 + (1+\alpha)\|X\|_F^2 \ge 4\alpha\|(u, X)\|_\star^2,$$

which proves (6.3). □

For the matching upper bound, take

$$u = e_t, \qquad X = \mathrm{diag}(0, \lambda I_d), \qquad \lambda := \frac{2\delta_{d,\sigma} R}{d}. \tag{6.4}$$

Then $2u + XZ = (2, \lambda Y)$ and

$$\|X\|_F^2 = d\lambda^2 = \frac{4\delta_{d,\sigma}^2 R^2}{d} \le C\frac{\sigma}{R} = C\alpha, \qquad \|(u, X)\|_\star^2 \ge 1. \tag{6.5}$$

Let $g(S) = S/r$. From (3.5), exactly,

$$(2, \lambda Y)^\top A(2, \lambda Y) = q(s)\lambda^2 \left(g(S) - \frac{d}{R}\right)^2 + p(s)\lambda^2 \frac{\sigma^2 S}{r^3}. \tag{6.6}$$

The factorization

$$g(S) - \frac{d}{R} = (S - d)\frac{\sigma^2 + rR}{rR(r + R)} \tag{6.7}$$

implies

$$\left|g(S) - \frac{d}{R}\right| \le \frac{|S - d|}{R}, \qquad \mathbb{E}\left(g(S) - \frac{d}{R}\right)^2 \le \frac{2d}{R^2} \le 2. \tag{6.8}$$

Also, by (4.7),

$$\mathbb{E}\frac{S}{r^3} \le \mathbb{E}\frac{1}{r} \le \frac{C}{R}. \tag{6.9}$$

Using $q \le (4\sigma)^{-1}$, $\delta_{d,\sigma}^2 \asymp \sigma/R$, and $c_{d,\sigma} \asymp R$, equations (6.6)–(6.9) give

$$c_{d,\sigma}\mathbb{E}[(2, \lambda Y)^\top A(2, \lambda Y)] \le C\left(\frac{1}{d} + \frac{\sigma^3 R}{d^2}\right) \le C\frac{\sigma}{R} = C\alpha. \tag{6.10}$$

Moreover

$$\alpha\mathbb{E}\|(2, \lambda Y)\|^2 = \alpha(4 + d\lambda^2) \le C\alpha. \tag{6.11}$$

Substitution into (6.2), followed by division by the Fisher norm, proves

$$\gamma_{\mathrm{loc}}(V_{d,\sigma}) \le C\frac{\sigma}{R}. \tag{6.12}$$

The alpha floor gives the reverse bound. Together with (4.12),

$$\gamma_{\mathrm{loc}}(V_{d,\sigma}) \asymp \frac{\sigma}{R} \asymp \kappa^{-1/2}. \tag{6.13}$$

7 First-Hermite certificate

Let

$$B := \mathrm{diag}(0, I_d/\sqrt{d}), \qquad \|B\|_F = 1. \tag{7.1}$$

Hessian compatibility and two Gaussian integrations by parts give

$$\begin{aligned} e_t^\top T[B] &= \mathbb{E}[t\,\mathrm{Tr}(BW_{d,\sigma})] = \mathbb{E}[(BZ)^\top W_{d,\sigma} e_t] \\ &= -\frac{c_{d,\sigma}\delta_{d,\sigma}}{\sqrt{d}}\mathbb{E}\left[q(s)\frac{S}{r}\right]. \end{aligned} \tag{7.2}$$

On the event $\mathcal{E}_d$ from Lemma 4.1, $q(s) \ge c/\sigma$ and $S/r \ge c\sqrt{d}$. Hence

$$\tau_H^2 \ge |e_t^\top T[B]|^2 \ge c\frac{c_{d,\sigma}^2 \delta_{d,\sigma}^2}{\sigma^2} \ge c\frac{R}{\sigma} \asymp \sqrt{\kappa}. \tag{7.3}$$

This explicitly realizes the central Frobenius trap rather than using it: the averaged t-derivative of the transverse Hessian has tiny individual eigenvalues but rank d, so its Frobenius norm is large even though the pointwise injective third-tensor norm is $L_H \asymp R/\sigma^2$.

Combining (4.12), (5.8), (6.13), and (7.3) yields the complete two-parameter frontier

$$\boxed{\kappa \asymp \frac{d}{\sigma^2}, \qquad L_H \asymp \frac{\sqrt{d}}{\sigma^2}, \qquad \gamma_{\mathrm{loc}} \asymp \frac{\sigma}{\sqrt{d}} \asymp \kappa^{-1/2}, \qquad \tau_H^2 \gtrsim \frac{\sqrt{d}}{\sigma} \asymp \sqrt{\kappa}.} \tag{7.4}$$

The specializations $(d, \sigma) = (n^4, n)$ and (n^6, n^2) prove Theorem 1.1. More generally, keeping $d \asymp \kappa\sigma^2$ leaves the square-root gap unchanged while $L_H \asymp \sqrt{\kappa}/\sigma$ can be made arbitrarily small by increasing the dimension.

8 Whitening transfer and guardrail audit

If an original-coordinate potential V^0 has Hessian-Lipschitz modulus L_H^0 and whitening uses $x = m_\star + \Sigma_\star^{1/2} z$, then

$$V^{\mathrm{w}}(z) = V^0(m_\star + \Sigma_\star^{1/2} z), \qquad \nabla^2 V^{\mathrm{w}}(z) = \Sigma_\star^{1/2} \nabla^2 V^0(x) \Sigma_\star^{1/2}.$$

Therefore

$$L_H^{\mathrm{w}} \le L_H^0 \|\Sigma_\star^{1/2}\|_{\mathrm{op}}^3. \tag{8.1}$$

The construction above requires no transfer: $\Sigma_\star = I$, and every $\alpha, \beta, \kappa, L_H$ is already a whitened quantity.

The proof meets the stated guardrails as follows.

1. Equation (3.5) is the pointwise Hessian of the displayed C^∞ potential. Full third- and fourth-order compatibility, not first-Hermite symmetry alone, is used in Lemma 6.1 and (7.2).
2. No commutation-free Loewner transfer occurs. The inverse gap is the scalar Rayleigh calculation (6.6)–(6.13).
3. Bounds and isotropic mean are not treated as sufficient abstract data. Exact isotropy comes from (4.5), and the small energy comes from the compatible radial and mixed blocks of (3.5).
4. Both κ and L_H are post-whitening quantities. The general coordinate conversion is recorded separately in (8.1).
5. The Hessian and every derivative of the Hessian are bounded, so every Gaussian integration by parts used here is classical. No unproved weak extension is needed.
6. The inverse gap is $\Theta(\sqrt{\kappa})$, genuinely larger than $O(\log \kappa)$, even when $L_H \to 0$.
7. No statement about a nonlinear attraction radius is made.
8. At no point is an operator bound on $\nabla^3 V$ converted into a Frobenius bound. The large Frobenius coupling is instead certified directly by the normalized test (7.1).